

1. First Flight of the Space Shuttle (Synchronization Problem)

- **Event:** Space Shuttle **STS-1** first launch attempt.
 - **Date:** **April 10, 1981** (launch scrubbed; actual first flight was April 12, 1981).
 - **Cause:** The Shuttle had **five redundant General Purpose Computers (GPCs)** that needed to be synchronized. A **synchronization issue** between them (caused by a countdown clock drift in one GPC) forced NASA to scrub the April 10 launch.
 - **Effect:** Launch was delayed 2 days. On April 12, STS-1 successfully launched.
 - **Lesson:** Even in redundant real-time systems, **synchronization across multiple computers** is critical.
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2. Ariane 5 (Overflow)

- **Event:** Ariane 5 Flight 501 failure.
 - **Date:** **June 4, 1996.**
 - **Cause:** A **numeric overflow** occurred when converting a 64-bit floating-point number to a 16-bit signed integer in the **Inertial Reference System (SRI)** software. The value was valid for Ariane 4 but not for Ariane 5's higher velocity.
 - **Effect:** The rocket veered off course ~37 seconds after launch and self-destructed.
 - **Lesson:** Reuse of software **without revalidation for new operational parameters** can be catastrophic.
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3. Airbus A320 (Cart Task Problem)

- **Event:** Crash of **Air France Flight 296Q** during a demonstration flight.
- **Date:** **June 26, 1988** at Mulhouse-Habsheim Air Show.
- **Cause:** The **Airbus A320's flight control system** (fly-by-wire) did not respond as expected during a low-pass flyover. Some reports point to the **slow execution of certain background (cartographic) tasks** in the Flight Management System, contributing to late throttle response and ground impact.

- **Effect:** Aircraft struck trees at the end of the runway, killing 3 passengers.
 - **Lesson:** Mismanagement of **task priorities** and human-machine interface design in real-time systems can have severe consequences.
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4. Mars Pathfinder (Reset due to Timeout)

- **Event:** NASA's Mars Pathfinder mission.
 - **Date:** July 4, 1997 (landing on Mars; problem occurred shortly after).
 - **Cause:** Pathfinder's computer experienced **frequent resets** due to a **priority inversion problem** in its VxWorks RTOS. A **high-priority task was waiting on a resource held by a low-priority task**, while a medium-priority task prevented the low-priority task from completing. This led to watchdog timeouts and system resets.
 - **Effect:** Initially appeared as a serious failure, but NASA engineers debugged it remotely and patched it by enabling **priority inheritance** in the RTOS.
 - **Lesson:** Classic textbook case of **priority inversion** in real-time systems.
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5. Malaysia Railway and Metro Errors

- **Malaysia (2021):** A **programming error** in the SelTrac signalling system caused the **Automatic Train Protection (ATP)** system to malfunction, contributing to a collision in Kuala Lumpur's Kelana Jaya LRT line. ([Beningo Embedded Group, Wikipedia](#))
- **China (2022):** At Qi'an Road station on Shanghai Metro Line 15, misoperation of the train-door control system allowed a passenger to be fatally caught due to door and screen-door misalignment triggered by system logic issues. ([Wikipedia](#))

These illustrate persistent risks in real-time control systems in mass transit.

✔ **Summary Table (Chronological Order):**

Event	Date	Cause (Software Issue)
Apollo 11 LEM Landing (Task Overrun)	July 20, 1969	Task overload from radar interrupts, handled by AGC
Space Shuttle STS-1 (Sync Issue)	April 10, 1981	GPC synchronization error, launch delayed
Airbus A320 (Cart Task Crash)	June 26, 1988	Task priority delays, throttle late response
Airbus A320 (Holding Task Late Issue)	1980s–1990s	Task scheduling / priority inversion
Ariane 5 Flight 501 (Overflow)	June 4, 1996	Integer overflow in reused inertial system software
Mars Pathfinder (Reset for Timeout)	July 1997	Priority inversion in VxWorks RTOS

Summary Table

Incident	Date	System Type	Cause / Notes
Talgo AVRIL trains (Spain)	Jan 1, 2025	High-speed train embedded SW	Date-processing bug in battery module halted operations
CrowdStrike kernel crash	Jul 19, 2024	Kernel-level security software	Faulty config update caused massive system crashes
SelTrac (Malaysia, 2021)	ATP 2021	Train signaling system	RT Programming error caused failure in automatic protection logic
Shanghai Metro door system (2022)	2022	Train door control system	Control logic misconfiguration led to fatal incident